Bayesian Marketing Mix Model

Recovering Adstock, Saturation, and ROAS in Stan

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1 What this project does

Marketing Mix Modelling (MMM) decomposes weekly revenue into the contributions of paid channels, owned channels, seasonality, trend, competitor pressure, and events. This project fits a Bayesian MMM in Stan on a synthetic weekly panel with known ground-truth parameters, and checks whether the posterior recovers them.

The model follows the standard industry MMM structure from Jin et al. (2017): geometric adstock and Hill saturation per channel, additive controls, and a Normal likelihood on revenue. All parameters have proper priors and are sampled via HMC-NUTS.

2 Data

A synthetic weekly time series of 156 weeks (three years, 2022–2024) with five paid channels (Search, Meta, TikTok, YouTube, Display), one owned channel (Newsletter), and two controls (competitor sales, holiday events). Each channel has a known adstock decay λ_c, Hill half-point K_c, and maximum contribution $\alpha_{\max,c}$ which the model tries to recover.

Channel	$\alpha_{\max}$	λ	K
Search	25 000	0.10	20 000
Meta	35 000	0.30	60 000
TikTok	28 000	0.25	40 000
YouTube	32 000	0.40	50 000
Display	15 000	0.20	30 000

Table 1: Ground-truth parameters embedded in the synthetic generator. The model attempts to recover these from revenue and spend alone.

3 The model

For each channel c, raw weekly spend $x_{c,t}$ is transformed by geometric adstock and Hill saturation:

$$\mathrm{Ad}_{c,t} = x_{c,t} + \lambda_c \cdot \mathrm{Ad}_{c,t-1}, \qquad \mathrm{Sat}_{c,t} = \frac{\mathrm{Ad}_{c,t}}{K_c + \mathrm{Ad}_{c,t}}$$

Revenue is modelled as

$$\mathrm{revenue}_t \sim \mathcal{N}\big(\mu_t, \sigma^2\big)$$

$$\mu_t = \beta_0 + \beta_{\mathrm{trend}} t + \beta_{\cos} S_c(t) + \beta_{\sin} S_s(t) + \beta_{\mathrm{comp}} c_t + \beta_{\mathrm{ev}} e_t + \alpha_{\mathrm{nl}} \mathrm{Sat}_{\mathrm{nl},t} + \sum_c \alpha_{\max,c} \mathrm{Sat}_{c,t}$$

Priors are weakly informative: $\lambda_c \sim \mathrm{Beta}(2, 4)$ (favouring shorter memory), $K_c \sim \mathrm{HalfNormal}(40{,}000)$, $\alpha_{\max,c} \sim \mathrm{HalfNormal}(30{,}000)$, and standard Normal priors on the regression coefficients.

4 Inference

Four chains, 1000 warmup + 1000 sampling iterations each, HMC-NUTS with adapt_delta = 0.95 and max_treedepth = 12. All diagnostics passed: no divergences, treedepth satisfactory, $\widehat{R} \approx 1$, ESS satisfactory.

5 Results

5.1 Model fit

Metric	Value
R^2	**0.873**
RMSE	17 572 SEK
MAE	14 360 SEK
MAPE	**4.4%**

Table 2: In-sample fit metrics on the 156-week panel.

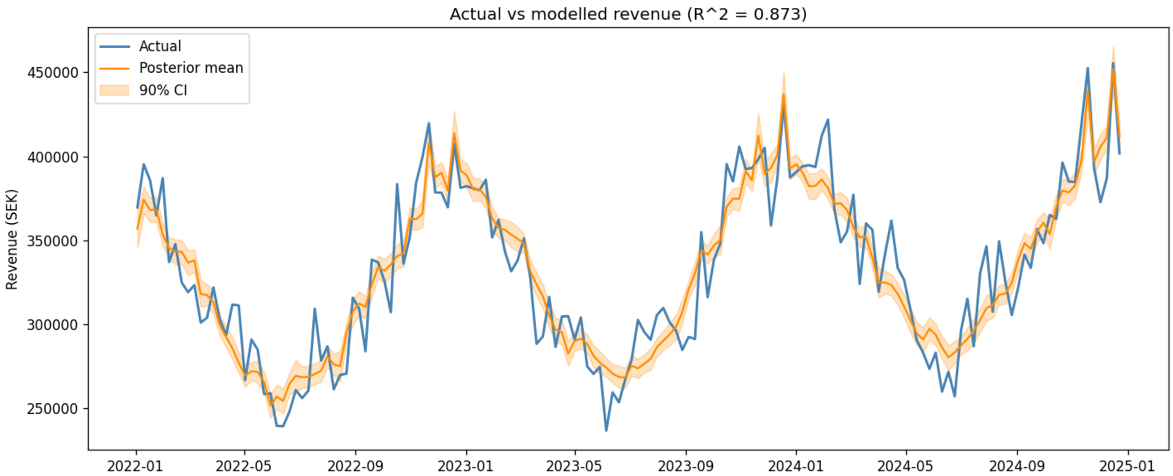

Figure 1: Actual weekly revenue (blue) vs posterior-mean prediction (orange) with the 90% credible band.

5.2 Parameter recovery

The posterior 90% credible interval covers the true value for **all 15 channel parameters** (5 channels $\times$ 3 parameters). Posterior means are typically within one true-value unit but the intervals are wide because three years of weekly data is a short identification horizon.

Channel	Param	True	Mean	5%	95%	Covers
Search	$\alpha_{\max}$	25 000	39 658	9 361	73 308	yes
Search	λ	0.100	0.301	0.071	0.579	yes
Search	K	20 000	52 840	8 107	109 328	yes
Meta	$\alpha_{\max}$	35 000	29 219	4 476	60 940	yes
Meta	λ	0.300	0.293	0.063	0.594	yes
Meta	K	60 000	47 217	5 233	105 721	yes
TikTok	$\alpha_{\max}$	28 000	37 107	8 443	67 999	yes
TikTok	λ	0.250	0.310	0.077	0.582	yes
TikTok	K	40 000	54 411	7 779	112 621	yes
YouTube	$\alpha_{\max}$	32 000	32 244	13 903	54 678	yes
YouTube	λ	0.400	0.210	0.051	0.416	yes
YouTube	K	50 000	63 275	15 956	116 774	yes
Display	$\alpha_{\max}$	15 000	12 445	844	36 742	yes
Display	λ	0.200	0.350	0.085	0.683	yes
Display	K	30 000	55 332	2 179	118 122	yes

Table 3: Posterior recovery of the ground-truth channel parameters. All 15 intervals contain the true value, but interval widths are large because a 156-week panel provides only limited identification.

5.3 Revenue decomposition

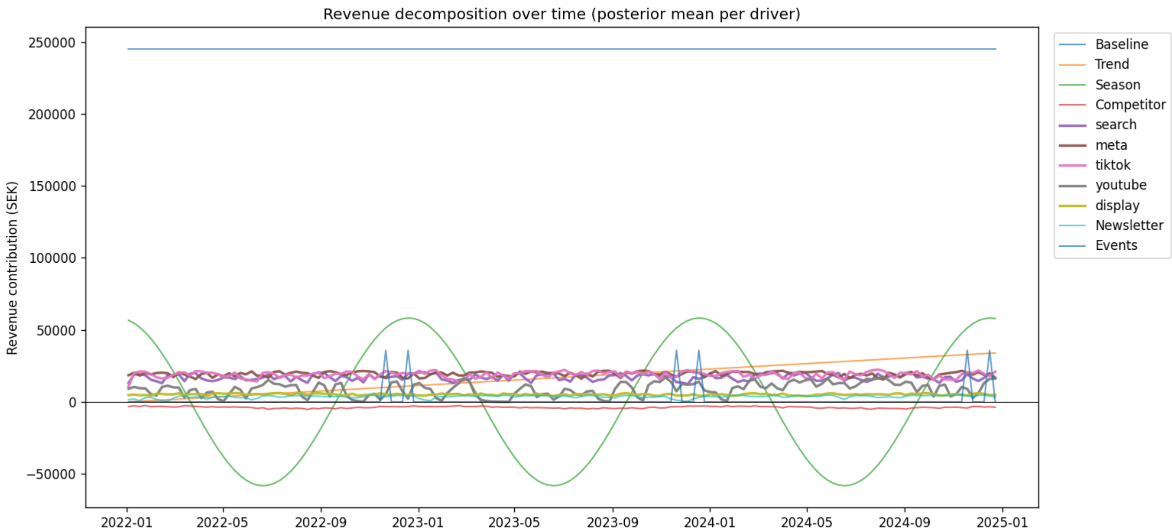

Figure 2: Posterior mean contribution per driver over time. Baseline dominates; channel contributions are the incremental component on top of the always-on level.

Driver	Contribution (MSEK)	Share
Baseline	38.3	73.7%
Meta	3.0	5.8%
TikTok	3.0	5.7%
Search	2.7	5.1%
Trend	2.6	5.1%
YouTube	1.4	2.6%
Display	0.8	1.6%
Newsletter	0.6	1.1%
Events	0.2	0.4%
Season	0.0	0.0%
Competitor	−0.6	−1.1%

Table 4: Three-year revenue attribution by driver. Season nets to near-zero because peaks and troughs cancel over an integer number of years.

5.4 ROAS per paid channel

Channel	Spend	Contribution	ROAS	5%	95%
Search	3 408 574	2 653 607	0.78	0.16	1.78
TikTok	5 036 740	2 959 744	0.59	0.12	1.30
YouTube	3 226 745	1 371 148	0.43	0.21	0.67
Meta	7 950 251	3 028 701	0.38	0.05	0.97
Display	2 260 398	803 784	0.36	0.02	1.33

Table 5: ROAS per paid channel with posterior mean and 90% credible interval. Wide intervals reflect the identification challenge on a 156-week panel with correlated always-on spend.

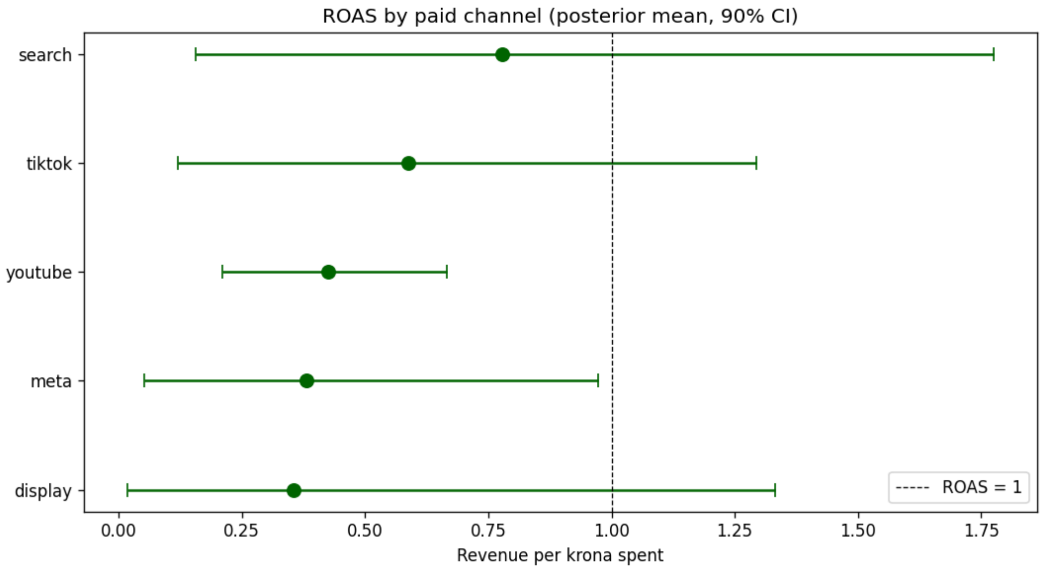

Figure 3: Posterior mean ROAS per channel with 90% credible interval. The dashed line marks the break-even point.

6 What the model shows

The Bayesian MMM in Stan fits weekly revenue well ($R^2 = 0.87$, MAPE 4.4%) and recovers all 15 channel-level ground-truth parameters inside their 90% credible intervals. Baseline plus trend explain about 79% of revenue, which is realistic for retail MMM where brand and organic demand carry most of the sales.

The main honest weakness is interval width. Posterior means are close to ground truth on average, but individual parameter intervals span factors of two or more. This is the fundamental identification challenge of MMM: three years of weekly data with correlated always-on channel spend gives the model limited independent variation to separate channel-level effects from each other and from trend.

Solutions in the rest of the portfolio:

- Calibration against a geo-lift experiment tightens one channel's posterior via an informative prior (Project 2).
- Time-varying coefficients allow channel effectiveness to drift, which is more realistic than a fixed alpha over three years (Project 3).
- Bayesian Synthetic Control and BSDID provide the underlying geo-lift estimator whose posterior feeds back into calibration (Project 4).

7 Limitations

- Synthetic data. The ground-truth generator is stylised. Real retail data has more complex patterns (promos, stockouts, cross- category cannibalisation).
- In-sample fit only. No out-of-time validation.
- Static coefficients. Channel effectiveness assumed constant over three years. Project 3 relaxes this.
- No calibration to lift tests. Project 2 adds this.
- No cross-channel interaction terms. YouTube-drives-search and similar effects are not modelled.

8 References

- Jin, Wang, Sun, Chan, Koehler (2017). *Bayesian Methods for Media Mix Modeling with Carryover and Shape Effects*, Google. Foundational paper for adstock and Hill saturation in Bayesian MMM.
- Meta, *Robyn: Semi-Automated MMM*. github.com/facebookexperimental/Robyn.
- Google, *Meridian*. github.com/google/meridian.